Introduction to Python programming

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Course material available on:
https://github.com/leops95/intro_to_python

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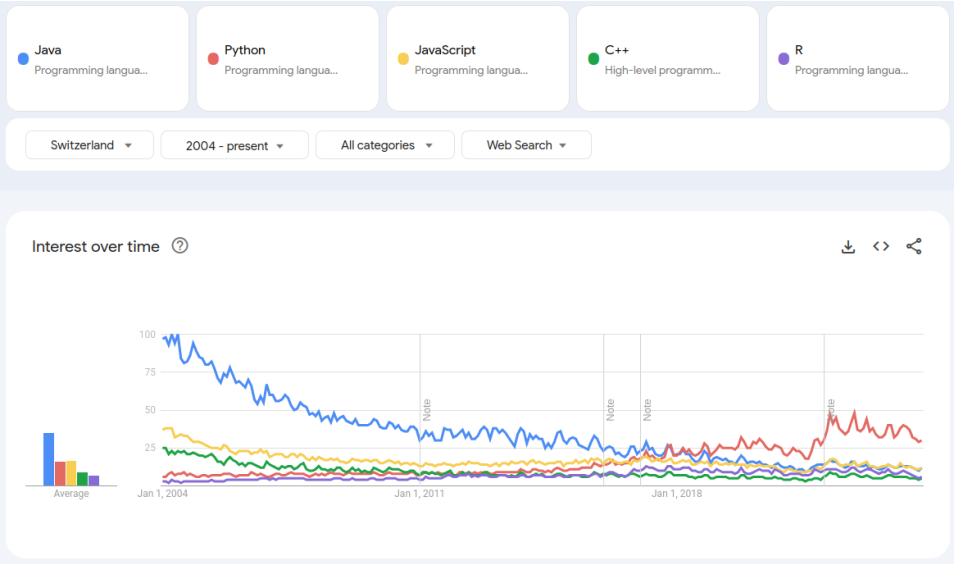
What's Python?

- General-purpose programming language
- Free and open source
- Elegant and user-friendly syntax
- Many useful libraries (Pandas, NumPy, Matplotlib, OpenCV, NLTK, statsmodels, Scikit-learn, PyTorch...)

One of the most popular programming languages



Learning curve

- The learning curve is hard at first
- It gets easier with experience:
  - Errors and syntax issues become easier to spot
  - You can use your past projects to help you
- No one knows everything by heart
- My goal is to show you the basics and help you to **become independent**

Objectives

1. **Set-up**: Install and use Python
2. **Python essentials**: The syntax, data types and basic operators
3. **Scientific computing**: Load datasets and work with them, plot data
4. **Becoming independent**: Troubleshooting your errors and ask for help online

Before we start

- This course is for **you**, I'm adapting to your needs
- Tell us a bit about yourself!
  - Have you ever used Python?
  - Why would you like to learn Python?
  - Do you have any other programming experience?

Sections

1 Set-up
- Installation
- Setting up your environment

2 Python essentials
- Basics
- Variables and data types
- Operators and conditions
- Loops
- Functions
- Exercises

3 Scientific computing
- Accessing files
- Packages
- Loading a dataset
- Summary statistics
- Data manipulation
- Plotting data
- Exercise

4 Becoming independent
- Where you can find help
- How to troubleshoot your errors
- StackOverflow or LLMs?

Set-up

Installation

- To install the "core Python package" you can go to `https://www.python.org/`
- As we want to use Python for scientific programming, you only have to install "Anaconda": `https://www.anaconda.com/`

→ Anaconda is a free distribution for Python which provides the core Python package, plus a convenient software and the most popular scientific libraries

- We write and compile code in files ("scripts") with the extension ".py"
  For example, `filename.py`

Setting up your environment

Definition

The **IDE (Integrated Development Environment)** is the software we're using to run Python scripts

Different IDEs for different needs:

- Very light: Jupyter Notebook, Google Colab... (for small projects, like problem sets or step-by-step tutorials)
- Intermediate: Spyder (includes a data viewer and advanced debugging tools)
- Heavy but efficient: VS Code, PyCharm... (for big projects and software engineering)

Setting up your environment

- We will use the Spyder IDE which comes with Anaconda
- Load it either from the Anaconda navigator or using the terminal
- Spyder is split into different sections ("panes") providing us with information or access to certain features. The most important are:
  - The editor
  - The console
  - The variable explorer and plots
- You can add, move or remove panes (see "View" $\rightarrow$ "Panes")

Python essentials

Basics

- Using hashtags (#), we take notes ("comments") directly into the code
- Enclosing lines within quotation marks (""") makes multi-line comments
- To display something on the console, we use the `print()` function
- I use the symbol > to show what the result looks like on the console

```
# the command below is likely going to be the first thing
# you try to write in any programming language
print("Hello world!")

> "Hello world!"
```

Note: Most IDEs have a color scheme to distinguish different elements of code

Variables

- **Variables** store data in our program's memory
- Using the assignment operator "=", we give them names and values
- Variables can take different data types: numbers, text, they could be binary, complex numbers, contain a tuple, a list, even a dictionary!
- The variable explorer shows you the type of all variables you have stored in memory

```
# assign values to variables
number_1 = 15
my_name = "Leo"
num_list = [2, 5]
```

Multiple assignment

You can assign multiple values to do different variables in one line

```
# assign values to variables
number_1 = 15
my_name = "Leo"
num_list = [2, 5]

# delete them
del number_1, my_name, num_list

# assign them again all at once
number_1, my_name, num_list = 15, "Leo", [2,5]
```

Numbers

- There are two different types of data representing numbers
  - **Integers** (`int`): whole numbers (0, 1, 2, 5001, -9999)
  - **Floats** (`float`): numbers with decimals (1.1, 2.64, 6.666666...)
- Python may dynamically change variable types if values are affected

```
number_1, number_2 = 1.99, 15
type(number_2)

> <class 'int'>

number_2 = number_1 + number_2
type(number_2)

> <class 'float'>
```

Strings

- A **string** (`str`) is a series of characters
- Anything inside single or double quotes are strings
  For example: `"My name is..."` or `'Python is fun!'`
- We can also nest single and double quotes
  For example: `'He said, "I love my dog."'`
- Using **F-strings**, we can enter ("forward") any variable value within a string

```
name, birth = "Léo", 1995
sent = f"Hi! My name is {name} and I'm {2026-birth} years old."

print(sent)

> "Hi! My name is Léo and I'm 30 years old."
```

Booleans

- A **boolean** (`bool`) is a data type that has two possible values (`True` or `False`)
- They are often used to keep track of conditions
- We also get them from doing logical comparisons (e.g., is `2 == 3` → `False`)

```
boolname = False
print(boolname)

> False

boolname = (5**2 == 25)
print(boolname)

> True
```

Lists

- A **list** (`list`) is a sequence of items ("elements") in a particular order
- You can modify any element by accessing its position in the list ("index")

Important

The index numbering in Python starts at 0, not 1 (sorry Matlab users!)

```
listname = [1,4,5,8]; print(listname[2])

> 5

listname[2] = 7; print(listname)

> [1,4,7,8]
```

Lists

- Lists are **mutable** (i.e., we can change the index of elements)
- The following table shows the most important list methods

Method	Description
`listname.append(i)`	Add the item `i` at the end of the list
`listname.insert(x,i)`	Insert the item `i` at the index `x`
`listname.pop(x)`	Remove the item at position `x` and return it
`listname.copy(x)`	Return a copy of the list
`listname.sort()`	Sort all the items in the list (increasing by default)

List operations

Example	Outcome
`a = [1,2]; a.append(3)`	`> a = [1,2,3]`
`a = [1,2]; a.insert(1,3)`	`> a = [1,3,2]`
`a = [1,2,3]; popped = a.pop(1)`	`> a = [1,3]; popped = 2`
`a = [1,2]; b = a.copy()`	`> a = [1,2]; b = [1,2]`
`a = [4,1,5,3]; b = a.copy();` `a.sort(); b.sort(reverse = True)`	`> a = [1, 3, 4, 5]; b = [5, 4, 3, 1]`

Slicing lists

We can select only some elements within a list ("slice") with: `listname[a:b]`

Important

Element a is always included, but **b is always excluded**

```
colors = ["red", "green", "blue", "yellow"], print(colors[1:3])

> ['green', 'blue']

print(colors[1:], colors[-1:]) # last 3 elements, last element

> ['green', 'blue', 'yellow'] ['yellow']
```

Dictionaries

- **Dictionaries** (`dict`) are used to store data in pairs (key + value)
- Values can be retrieved by their key (unique)
- Assigning values to a new key creates a new element

```
dictname = {"BS": "Basel Stadt", "GE": "Geneva", "TI": "Ticino"}
print(dictname["BS"])

> "Basel Stadt"

dictname["ZG"] = "Zug"
print(dictname)

> {'BS': 'Basel Stadt', 'GE': 'Geneva', 'TI': 'Ticino', 'ZG': 'Zug'}
```

Dictionaries

- Dictionaries (and lists) can be nested
- Nests can contain another data type

```
dictname = {"owners": ("Antonia", "Elda"),
            "pets": {"dogs": ("Charlie", "Razmotte", "Nemo"),
                     "cats": ("Zazie", "Peps", "Zélie")}}

print(dictname["pets"]["dogs"])

> ('Charlie', 'Razmotte', 'Nemo')
```

Arithmetic Operators

Operator	Description	Example	Result
`+`	Addition	`10 + 5`	`15`
`-`	Substraction	`30 - 20`	`10`
`*`	Multiplication	`2 * 5`	`10`
`/`	Division	`6 / 2`	`3.0`
`%`	Modulus	`10 % 4`	`2`
`**`	Exponent	`2 ** 3`	`8`
`//`	Floor division	`9 // 4`	`2`

Comparison Operators

Operator	Description	Example	Result
`==`	equal	`4 == 3`	`False`
`!=`	not equal	`4 != 3`	`True`
`>`	greater than	`6 > 10`	`False`
`<`	less than	`2 < 5`	`True`
`>=`	greater or equal	`8 >= 3`	`False`
`<=`	less than or equal	`5 <= 5`	`True`

Logical Operations

Let's assume two Boolean variables, `x = True` and `y = False`

Operation	Description	Example	Result
`or`	Returns `True` if at least one Boolean is true	`x or y`	`True`
`and`	Returns `True` if both Booleans are true	`x and y`	`False`
`not`	Returns the opposite of the Boolean	`not x`	`False`

Conditions

If statements (`if`) execute a piece of code only if a **condition** is satisfied (`True`)

```
x, y = 5, 10

if y < x:
    print("y smaller than x")
else:
    print("y greater than x")

> "y greater than x"
```

- the `else` block runs only if the condition is not satisfied (`False`)
- For more than two conditions, you can insert `elif` ("else if") before `else`
- Careful! you need to use consistent spacing ("indentation") to structure blocks

For loops

- Often, we want to perform the same task repeatedly
- **For statements** (`for`) iterate over items, in the index order
- Iterating does not make a copy of the sequence

```
numbers = [4,34,2]

for number in numbers:
    print(number + 1)

> 5
> 35
> 3
```

List comprehension

Using brackets usually is more efficient to iterate over elements in a list

```
listname = [1, 2, 3, 4, 5, 6]

listname = [x*x for x in listname] # this iterates over all elements
print(listname)

> [1, 4, 9, 16, 25, 36]

# we can add conditions to subset elements
listname = [x for x in listname if x%2 == 0]
print(listname)

> [4, 16, 36]
```

How many loops?

- The `range(a, b)` function generates arithmetic progressions
- As with lists, the last element (`b`) is excluded
- It is commonly used with `for` loops, to iterate over specific values
- You need to name the current item (below, `i`) if you want to use its value inside the loop

```
for i in range(1, 4):
    print("Loop number", i)
```

```
> Loop number 1
> Loop number 2
> Loop number 3
```

How many loops?

The `len()` function gives you the number of elements ("length") of a list

```
floats = [1.2, 2.343, 0.44]

for i in range(len(floats)):
    print(i, floats[i])

> 0 1.2
> 1 2.343
> 2 0.44

# another example with list comprehension
list_loop = [2*i for i in range(5)]
print(list_loop)

> [0, 2, 4, 6, 8]
```

While loops

- **While statements** (`while`) execute a task repeatedly *while* a condition is true
- You can also stop the loop using `break`

```
i = 1
while i < 10:
    print(i)
    if i == 4:
        break
    i += 1 # equivalent to i = i + 1
```

```
> 1
> 2
> 3
> 4
```

Functions

Definition

A **function** saves a specific task, to be executed upon calling its name

- It saves us time as we don't have to write the same code again
- We first need to **define** (`def`) a function, by giving it:
  - A name
  - A set of parameters in parentheses (optional)
  - A description (optional but recommended)
  - A set of instructions
- After the function is defined, we can call it with the required parameters (if any)

Functions

An example using the Fibonacci series:

```
def fib(n):
    """
    Print a Fibonacci series up to n
    """
    a, b = 0, 1
    while a < n:
        print(a, end = ' ')
        a, b = b, a + b

fib(10)

> 0 1 1 2 3 5 8
```

Functions

Functions can **show** (`return`) a result and store it (if assigned to do so)

```
def squared(array):
    # Find the square of each element in a vector
    output = []
    for elem in array:
        elem_squared = elem**2
        output.append(elem_squared)
    return output

n = [2, 5, 10]
n_new = squared(n)

print(n_new)

> [4, 25, 100]
```

Lambda expressions

Functions can be time-wise inefficient

For simple operations, use **lambda expressions**: `(lambda x: operation)(value)`

```
def simple_operation(x):
    x_new = x**2-1
    return x_new

n_new = simple_operation(10); print(n_new)

> 99 # four lines of code to get this result

# Same with lambda expression in one line
(lambda x: x**2-1)(10)

> 99
```

Now it's your turn!

Some exercises to practice:

1) Create two variables, then swap their values
2) Create a list containing the numbers 0 to 9, then invert it (9 to 0)
3) Write a function that returns the square of all odds or even numbers between 0 and 20

The file `solutions.py` contains the answers

Scientific computing

Paths

Definition

Your computer stores files in folders ("directories"), which can be accessed using **paths**. The latter comes in different formats depending on your operating system.

Let's take the Desktop:

Windows	`C:\Users\`**`username`**`\Desktop`
MacOS	`/Users/`**`username`**`/Desktop`
Linux (Ubuntu)	`/home/`**`username`**`/Desktop`

Simply replace "username" by your own session user name

Note: `~\Desktop` is also valid

Paths

- The Python console is always looking at one directory
- Not sure which one is it? Just type `pwd` ("print working directory") in the console
- Paths can be **absolute** or **relative**
  - Absolute paths refer to the entire path to your destination
  - Relative paths refer to paths *relative* to the current directory
- Changing directory is easy: either enter a new (absolute) path or go up/down the path tree (relative to the working directory)

Note: ".." refers to the parent directory (i.e., for going down the tree)

Packages

Definition

Packages are a collection of modules (list of Python files) that we can **import** into our script. They contain functions that serve a purpose, and are ready to be used.

- First, search a package name on the internet, find the command to install it
  - `https://pypi.org/`
  - `https://anaconda.org/conda-forge/`
- Then, paste the command on the terminal with a package manager:
  - **Pip**: the default one (`pip install pandas`)
  - **Conda**: the Anaconda version (`conda install -c conda-forge pandas`)

Packages

Installing new packages can be tedious:

- You need to use the terminal (with Bash commands) to install them
- They come in different versions, which can conflict with each other
- They need to be stored in a folder listed in the "$PATH" variable where Python will look for them

→ Anaconda manages all of this for you.

Otherwise, here are tutorials on using Bash commands **[CLICK HERE FOR LINK]** and managing the $PATH variable **[CLICK HERE FOR LINK]**

Packages

Finally, we import a package into our script using the keyword `import`

```
import numpy

# draw two random values (normally distributed)
print(numpy.random.randn(2))

> array([-1.0856306 ,  0.99734545])
```

- Subpackages (e.g. "numpy.random") only contain some functions
- Calling `import numpy.random` instead of `import numpy` saves memory!

Packages

- The keyword **as** gives a nickname to the package
- The keyword **from** calls only specific subpackages or functions

```
import numpy as np # "np" is a widely adopted alias
from numpy import cos, pi

print(np.sin(np.pi))

> 1.2246467991473532e-16

print(cos(pi)) # using "from" allows us to omit "np."

> -1.0
```

Some examples

- **NumPy**: Basic package for scientific computing. Very fast with mathematical and matrix operations. You can create "ndarrays" which are flexible, efficient and also faster than lists.
- **SciPy**: More advanced than Numpy (e.g. find the determinant or the inverse of a matrix, solve linear equations).
- **Matplotlib**: Plotting data, with complete control over the outline of graphs.
- **Pandas**: Loading datasets and data manipulation.
- **Scikit-learn**: Classification, clustering, basic machine learning

Some examples

- **Requests + BeautifulSoup**: Scraping data from websites
- **NLTK, Regex, Fuzzywuzzy**: Text and natural language processing (NLP)
- **OpenCV**: Images and computer vision (CV)
- **Statsmodels**: Statistical analysis and regressions
- **Tensorflow, Keras, PyTorch**: Advanced machine learning

Loading a dataset

- We will use the **Pandas package** to work with datasets
- Pandas can load most types of **structured data** (e.g., spreadsheets)
- First, find the appropriate command to your dataset type, for example:
  - `pd.read_excel(path_data)` for Excel files (.xlsx, .xls)
  - `pd.read_csv(path_data)` for comma-separated values (.csv)
  - `pd.read_stata(path_data)` for Stata datasets (.dta)
  - Tons of other options (e.g., .json, .pkl etc...)
- Then, simply replace `path_data` with the path leading to your dataset

Caution! There is no native option to read Rdata files: install `pyreadr` instead!

Loading a dataset

- Pandas comes with a special data type to handle datasets: **DataFrames**
- They are very popular for handling structured data
- It can do all the data cleaning operations, like:
  - Rename variables, replace or filter values
  - Concatenate, merge, group rows and columns
- Fast and efficient up to a few gigabytes of data (if you have 16Gb of RAM)
- If memory becomes scarce: look for alternatives like Dask, Modin, or Vaex (many other packages exist)

Loading a dataset

A short example, using my own research on metaphors:

```
import os # to navigate between paths
import pandas as pd

os.chdir("/home/username/Desktop/intro_to_python-master")

df = pd.read_csv("data_raw/Idaho_2022.csv")
```

- Here, we use `os.chdir()` to set the working directory
- We capture paths in string format, do not forget " or ' around them

Summary statistics

Before going any further:

- A DataFrame contains observations ("rows") and variables ("columns")
- The **dimensions** of the DataFrame can be seen in the data viewer
- Each column has its own data type, use `df.dtypes` in the console to see them all at once
- Columns usually contain special data types called objects (`object`)

Mea Culpa

When I speak, I tend to use both Stata and Python terminologies

Summary statistics

→ Let's have a look at the DataFrame we have opened...

- We access columns using brackets: `df["filename"]`
- We access rows using their index: `df.iloc[1]`
- Subsetting rows in a dataset works just like lists: `df.iloc[1:3]`

Summary statistics

Basic summary statistics functions:

Function	Description
`df.dtypes`	Show all data types
`df["metaphor_score"].mean()`	Display the mean of the variable
`df["metaphor_score"].std()`	Display the standard error
`df["metaphor_score"].max()`	Display the maximum value
`df["metaphor_score"].describe()`	Display N, mean, std, p10, median...
`df["arg1"].value_counts()`	Tabulate all values and frequencies
`df["speaker"].unique()`	Look for duplicates

See website to translate Stata commands into Python **[CLICK HERE FOR LINK]** and R commands into Python **[CLICK HERE FOR LINK]**

Data manipulation

Let's apply some basic data manipulation techniques...

```
# Drop the filename column
df = df.drop(columns = ["filename"])

# Rename the state column
df = df.rename(columns = {"st_name": "state"})

# Filter out bad metaphor scores
df = df[df["metaphor_score"] >= 0.7]

# Create a new metaphor column
df["metaphor"] = df["arg0"] + " " + df["arg1"]
```

Apply and map

You can add or change rows using `apply()` (with a function) or `map()`

```
# Recode the gender variable from int to str
def recode_gender(x):
    gender_str = ""
    if x == 1:
        gender_str = "Woman"
    else:
        gender_str = "Man"
    return gender_str

df["gender_str"] = df.apply(lambda x: recode_gender(x["gender"]),
                                                        axis = 1)
# Alternatively,
gender_map = {0: "Man", 1: "Woman"}
df["gender_str2"] = df["gender"].map(gender_map)
```

Append

You can join datasets based on columns ("append") with the function `pd.concat()`

```
import os
import glob # to store many file names
import pandas as pd

os.chdir("/home/username/Desktop/into_to_python-master")

files = glob.glob("data_raw/*_2022.csv") # star = "any"

df = pd.DataFrame() # creates an empty DataFrame

for file in files:
    data = pd.read_csv(file)
    df = pd.concat([df, data])
```

Note: `df.append(data)` is **deprecated** (i.e, it is not updated anymore!)

Merge

You can also join other information based on rows ("merge")

```
df_party = pd.read_csv("data_raw/political_party.csv")

df_merged = df.merge(df_party, on = "st_name", indicator = True,
                     how = "outer") # or "left", "right", "inner"

# print the output of the merge
print(df_merged['_merge'].value_counts())

> both          13186
> right_only        1
> left_only         0
> Name: _merge, dtype: int64
```

Merge

- Here, we are in a situation where one speaker belongs to one party
- But we have multiple rows for each speaker!
- We can enforce the type of merge using the "validate" parameter:
  - `1:1` = one-to-one
  - `m:1` = many-to-one / `1:m` = one-to-many
  - `m:m` = many-to-many

```
df_merged = df.merge(df_party, on = "st_name", indicator = True,
                     validate = "m:1") # or "many_to_one"
```

Note: The default value for the parameter "how" is "inner"

Collapse

Now, which political party employs the most metaphors?

We can answer this question by grouping the data (by party)

```
df_merged = df_merged[df_merged["metaphor_score"] >= 0.7]
df_merged["nb_metaphors"] = 1 # each row is counted as one metaphor

df_collapsed = df_merged.groupby(
                "party", as_index = False)["nb_metaphors"].sum()

print(df_collapsed)

>         party  nb_metaphors
> 0    Democrat           220
> 1  Republican           305
```

Pivot

Finally, we can rearrange rows and columns ("reshape", or "pivot") of the dataset with the following:

```
df_collapsed["statistic"] = "metaphor frequency"

df_wide = df_collapsed.pivot(index = "statistic",
                             columns = "party",
                             values = "nb_metaphors")

print(df_wide)

> party                Democrat  Republican
> statistic
> metaphor frequency        220         305
```

Note: `stack` and `unstack` are elegant substitutes

Plotting data

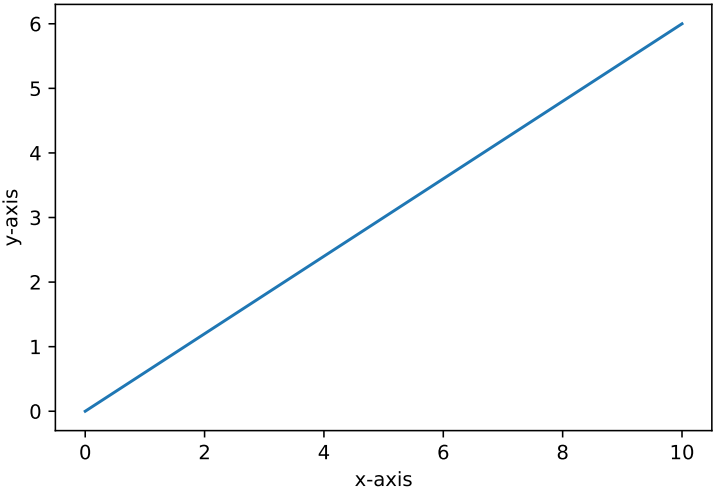
The easiest way to visualize ("plot") data is using the **Matplotlib** package

```
import matplotlib.pyplot as plt
import numpy as np

x_vals = np.linspace(0,10,10) # Generate 10 values within [0, 10]
y_vals = np.linspace(0,6,10)

plt.plot(x_vals, y_vals)
plt.ylabel("y-axis")
plt.xlabel("x-axis")
plt.savefig("plot_example.png") # save as png
plt.savefig("plot_example.pdf") # save as pdf
plt.show()
```

Plotting data



Plotting data

Useful Pyplot functions:

Function	Description
`plt.plot()`	Plot y versus x as lines and/or markers
`plt.ylabel()`	Set the label for the y-axis
`plt.xlabel()`	Set the label for the x-axis
`plt.axis()`	Method to get or set some axis properties
`plt.title()`	Set a title for the axes
`plt.scatter()`	A scatter plot of y vs x
`plt.bar()`	Make a bar plot
`plt.figure()`	Create a new figure
`plt.subtitle()`	Add a centered title to the figure
`plt.subplot()`	Add a subplot to the current figure
`plt.show()`	Display the figure

Plotting data

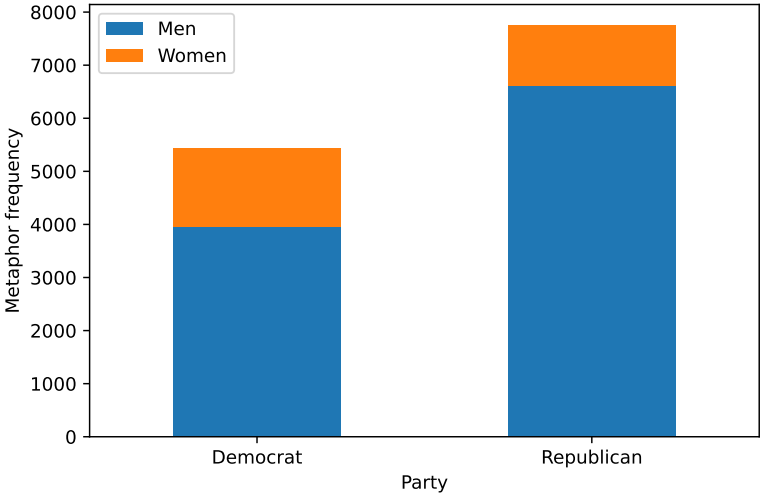
Histograms, pie charts, violin plots... everything is possible!

```
df_bar = df_merged.groupby(["party", "gender"],
                           as_index = False)["nb_metaphors"].sum()

df_bar = df_bar.pivot(index = "party",
                      columns = "gender", values = "nb_metaphors")

ax = df_bar.plot.bar(stacked = True, rot = 0)
ax.set_ylabel("Metaphor frequency"); ax.set_xlabel("Party")
ax.legend(["Men", "Women"])
plt.tight_layout()
plt.savefig("plot_example2.pdf")
plt.show()
```

Plotting data



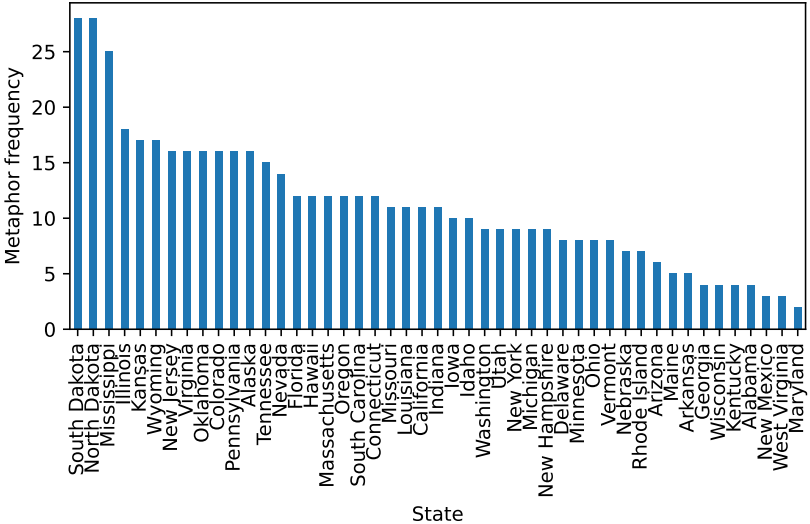
Note: Check out **Seaborn** for more customization options!

Now it's your turn!

Find out which U.S. state uses the most metaphors:

1. Append all datasets from the folder `data_raw`
2. Clean the columns of interest
3. Collapse the dataset to get the number of metaphors by state
4. Plot metaphor frequencies by state in a nice histogram

Example


Metaphor frequency
0
5
10
15
20
25
South Dakota
North Dakota
Mississippi
Illinois
Kansas
Wyoming
New Jersey
Virginia
Oklahoma
Colorado
Pennsylvania
Alaska
Tennessee
Nevada
Florida
Hawaii
Massachusetts
Oregon
South Carolina
Connecticut
Missouri
Louisiana
California
Indiana
Iowa
Idaho
Washington
Utah
New York
Michigan
New Hampshire
Delaware
Minnesota
Ohio
Vermont
Nebraska
Rhode Island
Arizona
Maine
Arkansas
Georgia
Wisconsin
Kentucky
Alabama
New Mexico
West Virginia
Maryland
State

Becoming independent

Where you can find help

The documentation: Every package comes with a document for each function, containing information on:

- What the function does
- The full list of arguments, what they mean, and their default value
- Some examples for using them

Websites of collaborative knowledge: StackOverflow (a few words on this later)

Where you can find help

Large language models (LLMs): great tools, but be careful of copy-pasting!

Search engines: Another way to find answers (tutorials, videos, short courses)

- Lot of content, although most of it does not answer your own *special* question
- Content is often outdated, or simply not the most efficient anymore

Friends and university staff: sharing your questions with someone also helps:

- Your peers may learn from your questions too
- ...but their time is limited!

How to troubleshoot your errors

Often we feel stuck because our script doesn't compile.

The console will tell you where an error occurred (line number and error type)

Who is the culprit?

- A syntax error:
  - Pay attention to indentation, mismatched parentheses or quote types
  - Verify the data type of intermediate values: some of them might be incompatible
- A package you imported:
  - Read its documentation
  - Check if you have the last version

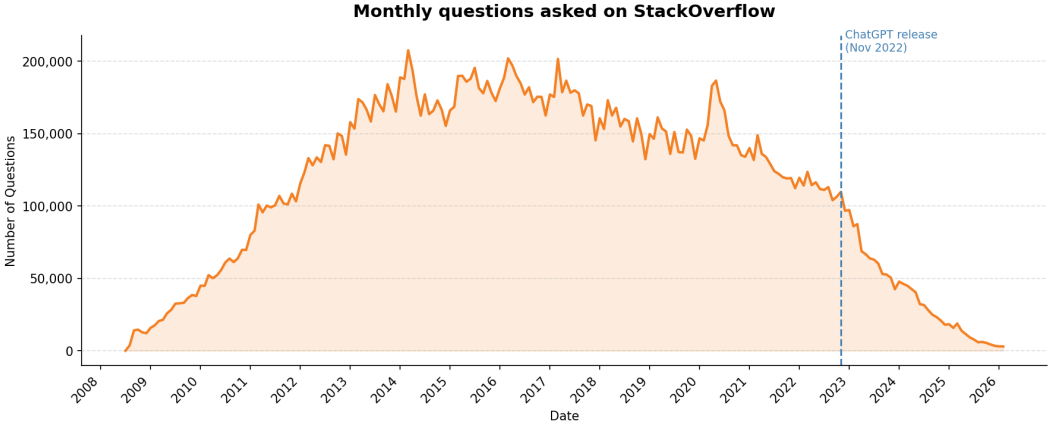
How to troubleshoot your errors

Who is the culprit? (continued)

- Something unclear within a function or loop I made:
  - Run a simpler version of your function/loop, then add one layer of complexity at a time
  - Add a print statement with intermediate values to find incorrect behavior
  - Make it run out of your script, then add it back when it does
- Something you copy-pasted online (StackOverflow or LLM)
  - Check how the problem was framed: missing information leads to bad answers $\rightarrow$ Incomplete representation of the data, issues of scale...
  - Understand every line before running it
  - Make it run out of your script, then add it back when it does

Is StackOverflow dead?

StackOverflow is a website whose goal is to capture all programming knowledge and make it available for others to learn.



Source: data.stackexchange.com

Using LLMs as a substitute

LLMs are becoming very accurate at answering general programming questions

StackOverflow or LLMs?

Nowadays, it is faster to ask an LLM for beginner programming questions. However, **we read faster than we write**: for intermediate/advanced questions, you may save some time by checking StackOverflow before typing a very long question for an LLM

In my opinion, learning with LLMs only works with **active focus**:

- Ask to explain each line of code the LLM produces
- Read and understand the code before running it: do not blindly copy-paste
- Take notes every time you learn something!

→ See Appendix slides if you wish to know more about using SO and LLMs more efficiently

Wrapping-up

Wrapping-up

With this course, you should now be able to:

- Install Python, set-up your first environment
- Understand most data types and work with them
- Load packages and datasets, perform basic data manipulation
- Efficiently look for help in the future...

Questions, remarks?

leo.picard@unibas.ch

Other fun stuff on: leopicard.net

Appendix: How to use StackOverflow efficiently

StackOverflow: How it works

- This website prioritizes quality over quantity of questions (or "posts")
- Do not ask a question before checking if it has already been answered before
- Only after, ask your question in the clearest and shortest way
  - Focus your question on what you don't know, skip all irrelevant details
  - Explain what you have tried before
  - Add a reproducible example (some code with fake data)
  - End your post by writing what the outcome should look like

→ Link to all the rules: `https://stackoverflow.com/help/how-to-ask`

StackOverflow: Badly written question

Python Script for column creation [closed]

Asked 3 days ago Modified 3 days ago Viewed 70 times

-5

Closed. This question needs details or clarify. It is not currently accepting answers.

Want to improve this question? As written, this question is lacking some of the information it needs to be answered. If the author adds details in comments, consider editing them into the question. Once there's sufficient detail to answer, vote to reopen the question.

Closed 3 days ago.

Improve this question

I have an excel file with no column names. Just the default A, B, C etc. that aren't technically a header. I have a total of 57 columns of data in the file. I am looking for a quick python Script that will take my XLSX file and create a column name for all the default columns with data in them.

So if like default a to become says column 1, or column A etc. I'm using python pandas. I am working on creating scripts to speed up some work.

I've already made a script that will take my .csv file with strip all columns except for the defined columns, however I am trying to take my original XLSX file and add column names/headers. I am brand new to any type of scripting and used this forum to manage one to extract specific columns.

I can open my original file, select every column and the data and create a table which then creates a column and header, but I'm trying to automate the process of opening the file, selecting all the data and creating a column name for each so that the next script I made will delete all the unneeded columns.

I hope this makes sense.

Nothing that's worked lol

python

StackOverflow: Nicely written question

how to skip 2 data index array on numpy

Asked 2 years, 7 months ago Modified 2 years, 7 months ago Viewed 92 times

1

so I have an 8x6 array, then I want to make the array 4x6.the data such as below:

```
array([[76, 34, 56, 32, 55, 66],
       [99, 23, 11, 34, 45, 32],
       [87, 98, 87, 23, 12, 77],
       [78, 98, 89, 28, 91, 72],
       [76, 42, 45, 23, 56, 87],
       [81, 22, 34, 42, 81, 23],
       [91, 23, 45, 67, 45, 34],
       [87, 98, 23, 45, 23, 55]])
```

the data I want such below:

```
array([[87, 98, 87, 23, 12, 77],
       [78, 98, 89, 28, 91, 72],
       [91, 23, 45, 67, 45, 34],
       [87, 98, 23, 45, 23, 55]])
```

I've try use the slices but it doesn't work for my case. the code is here:

```
import numpy as np

data = [
    [76,34,56,32,55,66],
    [99,23,11,34,45,32],
    [87,98,87,23,12,77],
    [78,98,89,28,91,72],
    [76,42,45,23,56,87],
    [81,22,34,42,81,23],
    [91,23,45,67,45,34],
    [87,98,23,45,23,55]
]

data = np.array(data)
data[::2]
```

StackOverflow: Careful!

- Response times are unpredictable: you could get an answer within minutes, but it can also take hours, even days... Sadly you may also wait for nothing :-(
- People won't always be nice to you (no need to say "hi" and "thanks" too)
- People might misunderstand your question, or tell you why you shouldn't do it this way
- People might give you a solution that works for the example you've laid out to them, but does not on your entire dataset (e.g., incomplete representation of the data, issues of scale...)

Appendix: How to use LLMs efficiently

LLMs: How it works

- LLMs (e.g., ChatGPT, Claude, Gemini, Le Chat) generate answers based on patterns learned during training. They do not "look up" the correct answer
- The quality of the answer depends heavily on the quality of your question ("prompt")
- Structure your prompt clearly:
  - Describe the context and what you are trying to achieve
  - Specify the language and any relevant libraries (e.g., "in Python using pandas")
  - Explain what you have already tried and what went wrong
  - Add a short reproducible example (provide some data in a .csv file)
  - State clearly what the expected output should look like

→ General rule: the more context you give, the better the answer

How to write a good prompt (1/2)

- **Be precise**
  - × *"Fix my code"*
  - ✓ *"I'm getting an* `IndexError: list index out of range` *on line 12. Here is my code: [...]. What is causing it?"*
- **Add relevant context**
  - × *"Why doesn't my loop work?"*
  - ✓ *"This* `for` *loop should print each word in a list but only prints the last one. Here is the code: [...]. What am I doing wrong?"*
- **Ask for understanding**
  - × *"Write me a function that sorts a list"*
  - ✓ *"Write me a function that sorts a list and explain each line so I can understand how it works"*

How to write a good prompt (2/2)

- **Ask for examples**
  - × *"Teach me `for` loops in Python"*
  - ✓ *"I'm a beginner in Python. Can you explain how `for` loops work with a simple example using a list of names?"*
- **Mention what you have tried**
  - × *"My script crashes when I read a CSV file"*
  - ✓ *"My script crashes with `FileNotFoundError` using `pd.read_csv('data.csv')`. The file is in Downloads and my script is on the Desktop. I think it's a path issue. How do I fix it?"*

LLMs: Careful!

- LLMs can be confidently wrong: they may give you code that looks correct but doesn't work, or silently solves a different problem than yours
- Always read and *understand* the code before running it: do not blindly copy-paste
- The solution may work on the example you gave, but fail on your actual data (e.g., different column names, missing values, issues of scale...)
- LLMs have a knowledge cutoff: they may be unaware of recent package updates or deprecated functions
- Each conversation starts fresh: if you start a new chat, the LLM has no memory of your previous exchanges

LLMs: To go further

With recent advancements in Agentic AI (OpenAI Codex, Claude Code, Gemini CLI, Mistral Vibe), you can let an LLM interact with files in your project environment

You benefit from saving time on repetitive tasks and avoid human errors, but:

- Options are generally not free
- There are obvious issues with data protection
- AI agents also make mistakes, including irreversible ones! (e.g., deleting important files)
- Also, there are rare risks of **prompt injection**, where malicious content in files or websites tricks the agent into doing something harmful

Scott Cunningham writes excellent guidelines on using Claude Code, check out his Substack: `https://causalinf.substack.com/s/claude-code`